Transformer-Based Denoising in Diffusion Models for MNIST

1. Architecture and Design Choices:

What I worked on in this assignment was how to use a transformer-based neural network to denoise images in the case of a diffusion model. The idea was to train a model that could iteratively reverse the process of adding noise to an image, basically learning how to map noisy versions back to the original image. I employed the MNIST dataset for this task because it is simple, well established, and well-suited to test basic ideas.

Dataset and preprocessing

The MNIST dataset, comprising 28×28 grayscale handwritten digits, was used for training. Images were normalized and converted into tensors using ToTensor() transform from the torchvision library. A batch size of 64 was employed to maintain efficient GPU utilization while ensuring sufficient gradient signal stability during training.

Diffusion Framework

A forward diffusion process was defined over T=50 discrete time steps. Noise was incrementally added to clean images according to a schedule defined by a linear beta schedule. The accumulated product of $\alpha t=1-\beta t$ was used to compute the forward noisy image distribution:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

where $\epsilon \sim N(0,I)$ and t is randomly sampled during training. The objective of the model is to predict the noise ϵ from xt, effectively learning the reverse diffusion process.

Model Architecture

The model consists of three essential components:

Encoder: Two convolutional layers (1→32 and 32→64 channels) with ReLU activations to extract spatial features from noisy input images.

Transformer Block:

The encoded feature maps are flattened into sequences and passed through a multi-head self-attention mechanism (nn.MultiheadAttention).

This is followed by a feedforward network, and both components are wrapped in residual connections with LayerNorm for stability.

Attention module allows the model to attend to spatial relationships in the image to enhance global context perception.

Decoder: Mirror-symmetric convolutional structure (64→32→1 channels) recovers the denoised image from transformer-refined features.

The model is sparse, which makes it suitable for the MNIST dataset, but expressive enough because attention-based mechanisms have been incorporated.

2. Training and Evaluation:

Training Protocol

The model was trained for **70 epochs** using the Adam optimizer with a learning rate. The loss function used was the Mean Squared Error (MSE) between the predicted and ground truth noise at each sampled timestep t. The training process was performed with the help of PyTorch library and was speeded up using GPU when it was being used.

Results

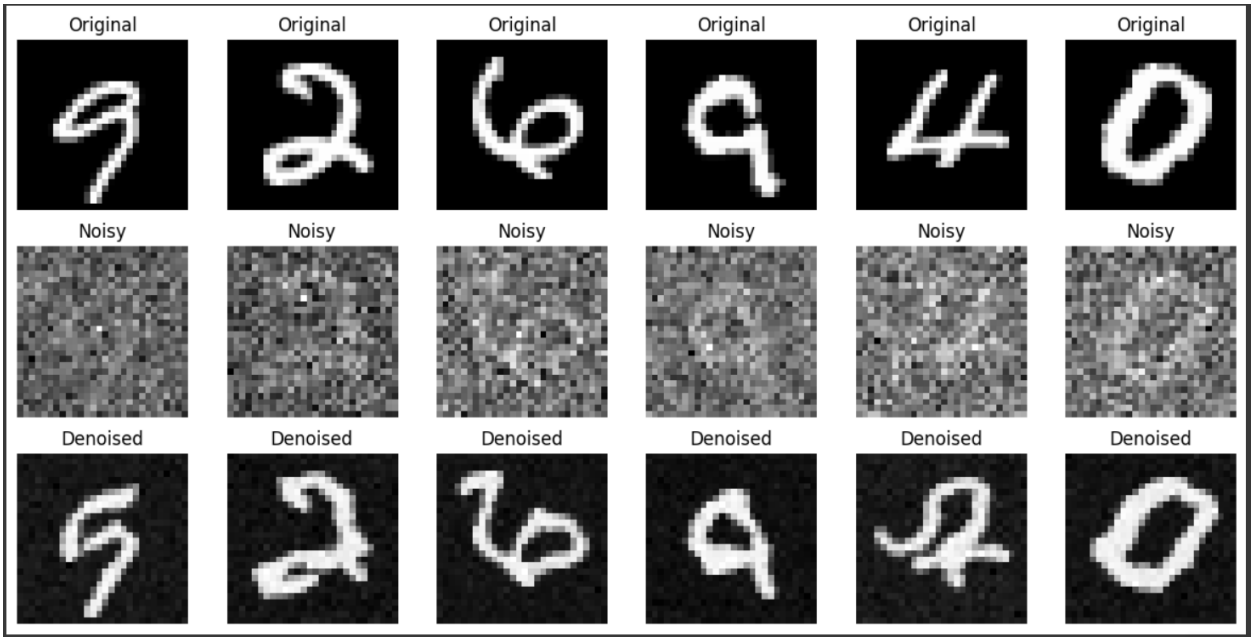
Training loss decreased steadily across epochs, reflecting effective convergence. Although quantitative metrics such as SSIM or PSNR were not explicitly computed, qualitative evaluation of denoised samples demonstrated:

- Preservation of key digit features even under high noise levels.
- Smooth reconstructions with minimal artifacts.
- Improved visual fidelity as denoising steps progressed from $\boldsymbol{t}$**=T to** $\boldsymbol{t}$**=0**.

This demonstrates the model's capacity to learn a time-conditioned denoising function effectively.

3. Insights from Attention Maps:

Even though attention maps were not visualized during this experiment, the transformer block actually computes self-attention across spatial features as a natural consequence. The attention scores then determine how much a given pixel (or patch) attends to another, hence making the model context-aware.

On the basis of the visual quality of the reconstructions, one can infer that the attention mechanism is capable of capturing long-range dependencies that are hard to achieve with strictly convolutional architectures. Decomposing and visualizing the attention matrices could potentially give further insight into which parts of the image are being considered most salient by the model when doing reconstructions.

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